EMIRCAN TUTAR

Software Engineer

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SUMMARY

Reliable, detail-oriented Software Engineer with around one year of professional experience in the automotive sector, focused on backend development, test automation, and CI/CD. Strong in Java, Python, and C++, with experience designing Docker-based, CI-ready infrastructure and automated pipelines with GitHub Actions, Jenkins, and Azure DevOps. Comfortable across the stack with REST APIs, SQL databases, and containerized distributed systems, and used to delivering well-tested results in agile Scrum teams. Backed by a portfolio of self-built projects spanning mobile, web, computer vision, and robotics. Passionate about agentic coding and AI-driven automation: currently building private projects with Claude Code and n8n, and actively exploring the latest developments in AI tooling and workflow automation. All projects: emirtutar.github.io/Portfolio/#/projects

SKILLS

Programming Languages:	Advanced: Java, Python, C++, SQL | Familiar: C#, JavaScript, HTML/CSS
Backend & Frameworks:	Spring, Java EE, REST APIs, Maven, Junit, CMake, eCAL, ROS
DevOps & CI/CD:	Git, Docker, Kubernetes, Jenkins, GitHub Actions, Azure DevOps, CI/CD
Databases:	SQL, Oracle, Firebase / Firestore (NoSQL)
Testing & Quality:	Unit, Integration, System, UI Testing, Robot Framework, pytest, gtest
Frontend:	Angular, React, Vue.js, Node.js
Tools & Methods:	Jira, Agile / Scrum, Agile Project Coordination
AI-Assisted Development:	Claude Code, ChatGPT, Gemini
Languages:	German (native), English (fluent), Turkish (native)

EXPERIENCE

Software Test Engineer (incl. Bachelor's Thesis)

Continental – ADC Automotive Distance Control Systems, Lindau · 03/2025 – 09/2025

Bachelor's Thesis: "Developing a Concept for Automated Testing of IPC Middleware – Demonstrated with eCAL"

- Built a fully automated test infrastructure for IPC/middleware that cut manual testing time by roughly 70% and made regression tests reproducible.
- Developed Docker- and Robot-Framework-based system and integration tests — isolated, parallelizable, and CI-ready.
- Designed around 40 test cases covering publish/subscribe, RPC, and crash and network-failure scenarios to safeguard critical communication paths.
- Integrated test execution into CI/CD pipelines (GitHub Actions), catching errors at commit time instead of late in the release cycle.

More details: *emirtutar.github.io/Portfolio/#/projects/ecal-test-suite*

Technologies: *Agile/Scrum, Java, Python, C++, Docker, Robot Framework, Git, GitHub Actions, CI/CD, eCAL, IPC / middleware, distributed systems, TCP/UDP, shared memory, test automation, test driven development*

Software Engineer – Software Bring-Up & Test (Internship Semester)

Continental – ADC Automotive Distance Control Systems, Lindau · 09/2024 – 03/2025

- Integrated software and hardware components for sensor data processing and inter-process communication (IPC).
- Developed and optimized C++ and Python modules, improving the runtime and maintainability of the processing pipeline.
- Applied eCAL middleware (publisher–subscriber model) and validated software in simulation and on real test drives.
- Automated build and test processes with Jenkins (CI/CD), largely eliminating manual builds.
- Led daily stand-up meetings for an 8-person development team when the Scrum Master was unavailable.

Technologies: *Jira, Agile/Scrum, Agile Project Coordination, C++, Python, Jenkins, CI/CD, eCAL, Git, system integration, sensor data processing, 3D simulation, middleware*

Java Programming Tutor

RWU – Ravensburg-Weingarten University: 09/2022 – 02/2023

- Supported students with practical Java lab exercises, debugging, and troubleshooting.
- Taught core programming and object-oriented design concepts and helped students structure their solutions.

Technologies: *Java, Spring, Maven, Object-Oriented Programming, Debugging*

Production and Assembly Assistant

ZF Friedrichshafen, Friedrichshafen: 09/2021 – 10/2021, 08/2022 – 09/2022, 08/2023 – 10/2023, 05/2026 – 08/2026

- Quality assurance of planetary carriers in series production; operated and maintained KUKA robots and press machines.

PROJECTS

RateMe - Android Product Rating App *(Team project)*

Barcode-scanning Android app with product lookup and a Firebase-backed rating system; email-verified authentication, a real-time Firestore backend, favourites, scan history, and a multi-user comment feed.

- ***More details:*** *emirtutar.github.io/Portfolio/#/projects/rateme*
- ***Technologies:*** *Android, Java, Firebase Auth, Firestore (NoSQL), REST API, Backend, Frontend, Git*

Patient Monitoring Prototype *(Team project)*

Camera-based fall and bed-exit detection using YOLOv5 on Raspberry Pi; MQTT messaging layer with encrypted staff alerts, packaged as a full Docker-Compose stack with hardware alarms (LEDs, buzzer).

- ***More details:*** *emirtutar.github.io/Portfolio/#/projects/patient-monitoring*
- ***Technologies:*** *Python, Docker, Raspberry Pi, YOLOv5, MQTT, Git*

AI-Assisted Software Testing

Evaluated GPT4o for automated test generation across unit, integration, and UI tests on three Java projects; measured code coverage, error rates, and prompt effort, documenting strengths and limits.

- ***More details:*** *emirtutar.github.io/Portfolio/#/projects/java-testing*
- ***Technologies:*** *Java, Maven, Spring, Junit, JavaFX, TestFX, GPT4o, Git*

Outdoor Planner – Weather-Aware Scheduler *(Team project)*

Vue.js single-page app combining live weather data (OpenWeatherMap) with an outdoor appointment manager; responsive layout and full appointment CRUD. Live at outdoorplaner.netlify.app.

- ***More details:*** emirtutar.github.io/Portfolio/#/projects/outdoor-planner
- ***Technologies:*** *Vue.js, JavaScript, REST API, html, CSS, Git*

Mission Fried Chicken *(Team project)*

Arcade game built to practise clean game architecture (game logic, UI, scene/menu structure, simple animations) with a focus on responsive controls and timing; implements the full game loop.

- ***More details:*** *https://emirtutar.github.io/Portfolio/#/projects/mission-fried-chicken*
- ***Technologies***: *C#, Azure DevOps (CI/CD build & test pipeline), Git*

EDUCATION

B.Sc. Applied Computer Science *(09/2021 – 03/2026)*
Ravensburg Weingarten University, Grade: 2.3
Specialization: Robotics and Smart Devices

Fachabitur *(09/2019 – 08/2021)*
Fachoberschule, Lindau (Bavaria), Grade: 2.4
Six-month school-based internship in metalworking, woodworking, and electrical work

Mittlere Reife *(09/2013 – 08/2019)*
Realschule im Dreiländereck, Lindau (Bavaria), Grade: 2.5
Elected student representative in years 9 and 10

Notes

Availability: Immediately and open to relocation.